



Tertiary Infotech Academy Pte Ltd

UEN: 201200696W

LESSON PLAN

For

Certified Lean Six Sigma White Belt (CLSSWB)

Course Code: C142

Conducted by

Tertiary Infotech Academy Pte Ltd

UEN: 201200696W

Version 2

DOCUMENT VERSION CONTROL RECORD

Version Number	Effective Date of Release	Summary of Included Changes	Author
1	1 July 2026	Initial release — CLSSWB 1-day lesson plan for the Lean Six Sigma awareness course.	Dr. Alfred Ang
2	19 July 2026	Lesson plan rebuilt from the single-source content module; schedule retimed to 9:30am – 5:30pm.	Dr. Alfred Ang

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Course Information

Course Title	Certified Lean Six Sigma White Belt (CLSSWB)
Course Reference	C142
Training Provider	Tertiary Infotech Academy Pte Ltd (UEN 201200696W)
Duration	1 day · 7.5 instructional hours
Daily Timing	9:30 am – 5:30 pm (30-minute lunch)
Mode	Instructor-led, hands-on Lean Six Sigma labs using the BrewBean Cafe morning rush improvement scenario
Trainer	Dr. Alfred Ang

Learning Outcomes

On completion of this course, learners will be able to:

- LO1: Describe the core concepts of quality, Lean, Six Sigma and Lean Six Sigma, and the belt roles in an improvement team.
- LO2: Explain the five phases of the DMAIC roadmap and what happens in each.
- LO3: Identify customer requirements and a clear problem statement in the Define phase.
- LO4: Describe how a process is mapped and measured, and identify the eight wastes in the Measure phase.
- LO5: Identify likely causes of a problem using 5 Whys and Fishbone analysis in the Analyze phase.
- LO6: Describe common improvement and control actions that fix a cause and hold the gain.

Course Schedule

Day 1 · 9:30am – 5:30pm

Time	Duration	Topic / Activity
9:30–10:00	30 min	Welcome, course introduction, ground rules and course portal walkthrough [slides 1–7]
10:00–11:05	65 min	FOUNDATIONS — What is Quality; What is Lean; What is Six Sigma; Lean vs Six Sigma vs Lean Six Sigma; the belt pathway; the DMAIC roadmap [slides 8–18]
11:05–11:55	50 min	DMAIC · DEFINE — Voice of the Customer; VOC to CTQ translation; problem

		statements; the project charter; SIPOC [slides 19–30]
11:55–12:30	35 min	Hands-on: Lab 1: Define — Customer Requirements and the Problem Statement [slides 31–37]
12:30–13:00	30 min	Lunch break
13:00–13:50	50 min	DMAIC · MEASURE — process mapping; SIPOC; types of data; data collection plans; check sheets; the eight wastes [slides 38–48]
13:50–14:25	35 min	Hands-on: Lab 2: Measure — Map the Process and Spot the Waste [slides 49–55]
14:25–15:15	50 min	DMAIC · ANALYZE — symptom vs root cause; 5 Whys; Fishbone (5M); reading a Pareto chart; common vs special cause variation [slides 56–67]
15:15–15:50	35 min	Hands-on: Lab 3: Analyze — Find the Root Cause using the 5 Whys and Fishbone [slides 68–74]
15:50–16:20	30 min	DMAIC · IMPROVE — generating solutions; impact/effort screening; 5S; mistake proofing; standard work; piloting [slides 75–85]
16:20–16:45	25 min	Hands-on: Lab 4: Improve — Choose and Pilot a Countermeasure [slides 86–92]
16:45–17:05	20 min	DMAIC · CONTROL — the control plan; visual management; SOPs; team huddles; project handover [slides 93–101]
17:05–17:20	15 min	Hands-on: Lab 5: Control — Hold the Gain and Hand Over [slides 102–109]
17:20–17:30	10 min	Course recap, next steps and course feedback [slides 110–118]

Day total: 7.5 instructional hours (9:30am – 5:30pm).

Lab Reference (aligned to the DMAIC phases)

DMAIC phase / Topic	Weighting	Labs
FOUNDATIONS: Lean Six Sigma Foundations	20%	
DEFINE: Define — Understand the Problem	20%	Lab 1
MEASURE: Measure — See What Is Really Happening	20%	Lab 2
ANALYZE: Analyze — Find the Cause	20%	Lab 3
IMPROVE: Improve — Fix the Cause	10%	Lab 4
CONTROL: Control — Hold the Gain	10%	Lab 5